

MetAgent case study

compound_only_enrich_mammalian_RAMP_P_000000016_seed1

Input metabolite set -> pathway evidence -> final metabolic interpretation

Input

Differential metabolites (n=10)

Ornithine

C5H12N2O2

C00077

2-Ketobutyric acid

C4H6O3

C00109

(R)-Lipoic acid

C8H14O2S2

C16241

S-Adenosylhomocysteine

C14H20N6O5S

C00021

L-Cysteine

C3H7NO2S

C00097

L-Methionine

C5H11NO2S

C00073

Other differential metabolites

1-Pyrroline-5-carboxylic acid (C03912),
5,6-Dihydroxyindole (C05578), Nitrate (C00244),
Phenylephrine (C07441)

Top pathway

Glycine, serine and threonine metabolism

KEGG map00260 | FDR=2.91e-13
fold-enrichment=568.9 | matched 6/10

Final verifier totals

supported 13 | unsupported 3 | contradicted 1
unverifiable_v0 14

Final MetAgent report

Verifier-colored complete narrative

supported

unsupported

contradicted

unverifiable_v0

The enriched metabolite set strongly implicates Glycine, serine and threonine metabolism (KEGG map00260) as the primary affected pathway (FDR = 0.0, fold enrichment = 568.9).

Six input metabolites are matched to this pathway: ornithine (C00077), 2-ketobutyric acid (C00109), lipoic acid (C16241), S-adenosylhomocysteine (C00021), L-cysteine (C00097), and L-methionine (C00073).

The biological story centers on interconnected amino-acid metabolism.

L-Methionine is upstream of S-Adenosylhomocysteine (SAH) in the methionine cycle (KEGG shortest path: methionine -> S-adenosylmethionine -> SAH; 2 hops), placing SAH elevation downstream of increased methionine flux.

L-Cysteine participates in transsulfuration, linking to the ornithine/proline axis through bidirectional connections to 1-Pyrroline-5-carboxylate.

Lipoic acid (C16241) is a cofactor forpyruvate dehydrogenase and 2-oxoglutarate dehydrogenase.

2-Ketobutyric acid (C00109) links serine/threonine catabolism to the citric acid cycle.

The simultaneous perturbation of ornithine, cysteine, methionine, and SAH is consistent with altered flux through Glycine, serine and threonine metabolism [retracted: Cysteine and methionine metabolism claim not supported by query_ramp_enrichment].

Limitations: The query_ramp_enrichment output returned only one top pathway; claims about additional pathways or their disease-state associations cannot be made from this result alone.